

```
exercise1.py
1  from sklearn.feature_extraction.text import CountVectorizer
2
3  # Collect user input
4  documents = []
5  text = input("Enter document contents or type `exit` to end input: ")
6  while text.strip() != 'exit':
7      documents.append(text.strip())
8      text = input("Enter document contents or type `exit` to end input: ")
9
10 # WRITE YOUR CODE HERE
11 #vectorizer = CountVectorizer(analyzer='word', ngram_range=(2,2), stop_words='english', l
12 #X = vectorizer.fit_transform(documents)
13 #print(vectorizer.get_feature_names_out())
14 #print(X.toarray())
15
16 vectorizer = CountVectorizer(analyzer='word', ngram_range=(2, 2), stop_words='english')
17 X = vectorizer.fit_transform(documents)
```

```
64% (12:41)
Terminal
codio@horizonliquid-elasticsusan:~/workspace$ python3 exercise1.py
Enter document contents or type `exit` to end input: This is the first document.
Enter document contents or type `exit` to end input: This document is the second document.
Enter document contents or type `exit` to end input: And this is the third one.
Enter document contents or type `exit` to end input: Is this the first document?
Enter document contents or type `exit` to end input: exit
['document second', 'second document']
[[0 0]
 [1 1]
 [0 0]
 [0 0]]
codio@horizonliquid-elasticsusan:~/workspace$
```

Guide

Collapse

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Coding Exercise 1

In the top-left pane, implement a bag of words model using `documents` which will capture keyboard input.

When creating your bag of words model, make sure to:

- Use `CountVectorizer`
- Create 2-grams
- Filter out punctuation
- Filter out stop words
- Ignore case
- Score by count

Output the (1) vocabulary and the (2) vectors.

Test your code using the terminal in the bottom-left:

TRY IT

Example Input/Output:

This is the first document.
This document is the second document.
And this is the third one.
Is this the first document?
exit

['document second', 'second document']
[[0 0]
 [1 1]
 [0 0]
 [0 0]]

Submit your code for feedback by pressing the button below.

The solution is:

vectorizer = CountVectorizer(analyzer='word', ngram_range=(2, 2), stop_w
X = vectorizer.fit_transform(documents)
print(vectorizer.get_feature_names())
print(X.toarray())

By default, `CountVectorizer` :

- Filters out punctuation
- Ignores case
- Scores by count

The two options that need to be overridden are `ngram_range` and `stop_words` :

vectorizer = CountVectorizer(analyzer='word', ngram_range=(2, 2), stop_w

This creates n-grams of size 2 and filters out stop words.

Check It!

LAST RUN on 12/27/2023, 4:40:42 AM

Check 1 passed
Check 2 passed
Check 3 passed

Next